PARTNER PRODUCT · FINTECH / AI IN FINANCE

DALIL

Evidence-grounded AML investigation copilot

ML risk assessment · evidence-based template · optional experimental local LLM

`SC-01..06` `[E1] -> evidence` `sha256 verify` `TEMPLATE MODE`

Live demo: **https://cornbread-left-licking.ngrok-free.dev**

Proposed owner: MKBANK FCC alert-investigation desk · partnership not confirmed

The problem: alerts nobody has time to read

40

alerts per analyst a day (assumption)

30-40 min

per alert (assumption, not measured)

57%

false positives — rules-only layer on our own synthetic benchmark*

Why not just call a cloud LLM?

- Production constraint: customer data stays inside the bank
- Analysts need transaction-level evidence for their conclusions
- An LLM citation alone does not establish that a claim is true

Rules don't just drown analysts — they still miss 20.9% of novel laundering variants (our synthetic novel split).

* measured on our held-out synthetic test set (novel split: 58.1%) — a baseline for the pain, not a bank statistic.

DALIL: one screen from alert to decision

1 0-100 score

Calibrated GBM (70%) + transparent scenario rules (30%), banded low / medium / high / critical

2 Typology hits

SC-01 structuring .. SC-06 turnover mismatch — each hit linked to its transactions

3 Evidence note

Template note built only from evidence [E#]; optional experimental local-LLM draft

4 Machine checks

[E#] exists · numbers & dates match cited evidence · meaning not checked — warnings

5 Attribution

Leave-one-out: which features moved the score, with sign and magnitude

6 Audit + export

Decision bound to analysis_id + input hash; server-built package, per-section SHA-256

One host, local inference — no external AI API

single host (demo: laptop)

CSV / alert queue
jury input or demo case

→ **Ingest**
strict: bad rows rejected

→ **Entity resolution**
ambiguity-guarded merges

→ **49 features**
analyst-readable names

→ **SC-01..06 + GBM**
rules 30% · model 70%

→ **Ensemble score**
band 0-100

Evidence pack
E1..En from features + txs

→ **llama.cpp :18081**
optional experimental draft

→ **Note checks**
IDs · numbers · dates (warn)

→ **Decision log**
analysis_id + hashes

→ **Export bundle**
canonical JSON + sha-256

SSE stage stream -> React SPA: queue / investigate / system — stages stream live while the pipeline runs

No external AI API · outbound traffic not audited · the ngrok demo runs on the presenter's laptop with synthetic data — a bank rollout needs its own deployment, security review and model-risk validation.

LIVE DEMO

100 seconds — real UI, real pipeline, no edits

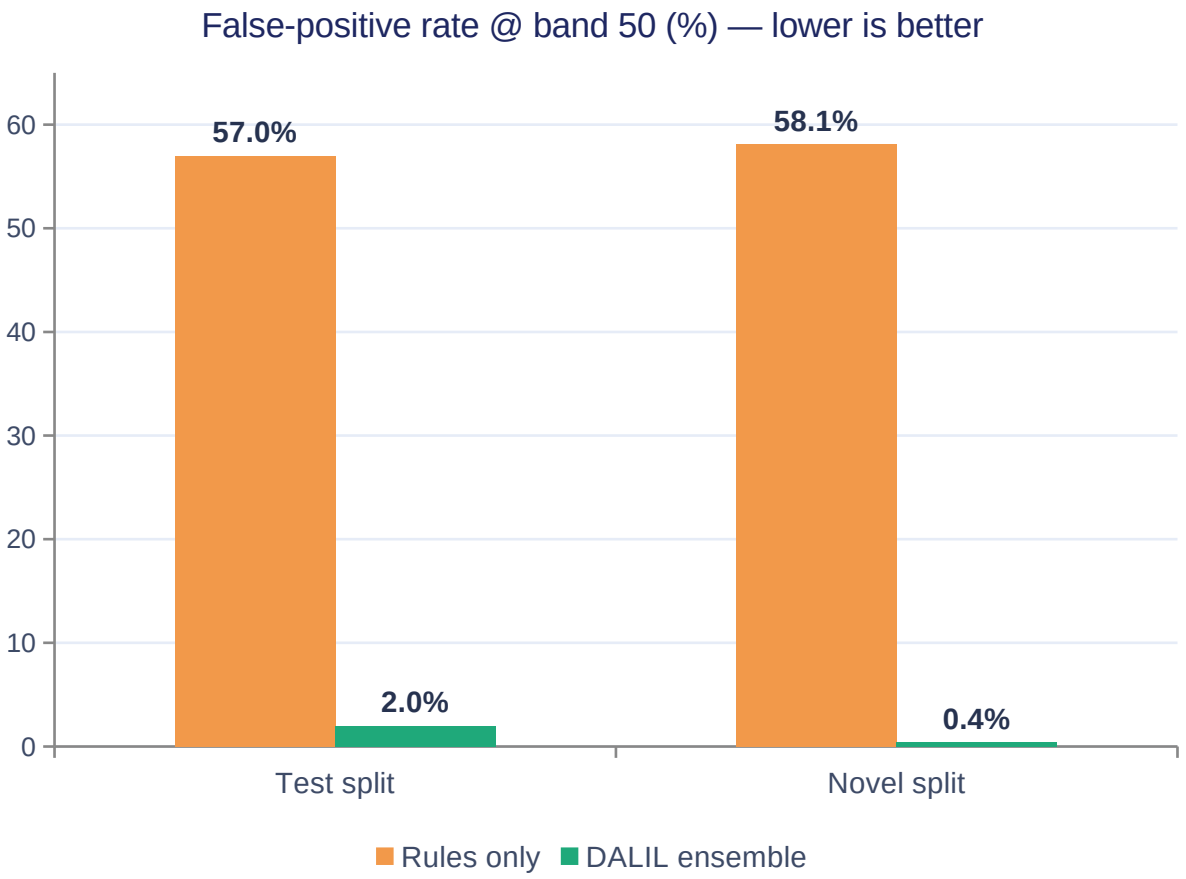
1. New CSV -> load template (12 synthetic txs, no customer metadata) -> Run
2. Score 83 critical · SC-01 hit · SC-03/SC-06 NOT EVALUATED · PROVISIONAL
3. Click a citation [E#] -> source transaction highlighted and scrolled into view
4. Separate checks: IDs exist / numbers & dates match / meaning NOT machine-verified
5. Name + justification -> Escalate -> reload (from server) -> Export: integrity + server match

`tunnel down -> localhost:8012`

`LLM down -> template note only`

`edited file -> scripts/verify_export.py (CLI)`

Honest numbers — synthetic, held-out, reproducible



99.2%

ensemble recall @ novel split, with 0.4% FPR — held-out synthetic variants

96.4%

precision @ 10% ASSUMED prevalence (novel) — a planning assumption, not bank data

0/20

Historical LLM drafts, before DEMO labels: zero check issues; numeric consistency 20%; meaning not measured

189 ms / 19 s

Historical p50 before DEMO labels: pipeline / LLM draft, 4 CPU threads

ROC-AUC 0.9945 / 0.9937 (test / novel) is optimistic because the data is synthetic. Failure list (5 FPs, 1 FN) ships in the UI. Full report: eval/results/eval_report.json

What is checked — and what is not

Citations exist in the evidence pack

Every [E#] must resolve to a real evidence id; a note with zero citations is reported as 0, not 'N/A'

Numbers & dates match cited evidence

Amounts (incl. 'mln'), counts and dates must match the evidence cited in that sentence — else flagged. Warnings, not a block.

Semantics NOT machine-verified

A valid citation does not make a claim true — the analyst reviews meaning; the LLM draft is experimental

Verifiable export

One JSON package:

input + SHA-256 · analysis_id · model/ruleset ids · results · evidence · note + LLM draft · decision

Built by the server from its own record; integrity (hashes) and server match are checked separately.

Limits printed inside the file: a hash is NOT a signature; an attacker holding the whole package can rewrite content and hashes together; LLM text is not bit-reproducible.

Local fit and validation limits

What ships in the repo

- Synthetic casebook: 3 080 cases, CC0, generator + DATA_CARD committed
- Code: MIT · model: Qwen2.5-1.5B (Apache-2.0) · llama.cpp (MIT)
- Ground-truth labels exist only offline — the API strips them from /api/queue
- 91 pytest tests + browser E2E: strict input, RU/UZ ambiguity, LLM down/truncated, decision binding, forged results, export tamper

Differentiation to validate

- UZS amount validation and RU/UZ name handling in one review workflow
- Local inference, linked transaction evidence and server-bound decisions
- Existing AML suites also manage cases; our small prototype has no proven barrier to copying
- Next evidence: analyst task-time study and an independent bank holdout; no confirmed partnership

Pilot budget and payback: illustrative assumptions

$6,000

One-off pilot: labour 200 h x $25 = $5,000; infrastructure $200; data review $500; reserve $300.

$2,000 / mo

Modelled net benefit: 4,000 alerts x 5 min / 60 x $15/h x 50% realised = $2,500; less $500/month operations.

3 months

Simple payback after rollout: $6,000 / $2,000. At 2,000 alerts/month: $750 net, 8 months. No measured savings.

Proposed 4-week workflow pilot

- Start with 50 authorised cases inside the bank; no real customer data in this demo
- Measure review time, corrections and missed evidence; this sample cannot establish a low FPR
  - Use a larger independent holdout for detection quality and recalibration
  - Budget and payback are assumptions, not a quote or bank agreement; security, data mapping and model validation remain required

The analyst stays in the loop.

No external AI API. Synthetic demo data.

Live: **https://cornbread-left-licking.ngrok-free.dev** Repo: github.com/funstorm15/dalil · README + eval report + 91 tests

Thank you — questions?

synthetic demo · MKBANK: proposed pilot, not a confirmed partnership · metrics are our own benchmark — see README §5